Test & Benchmark Report: Two-Stage Recommendation Engine

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Summary

The two-stage recommendation engine passes all 19 automated tests across 7 test files. Offline evaluation on a chronological held-out test set shows Stage 2 re-ranking with Wide & Deep improves NDCG@10 and Hit Rate@10 over retrieval alone. DCN-v2 scored higher on validation ROC-AUC but lower on top-rank ranking metrics, which is discussed in the findings below. Measured serving latency is well inside the production SLA targets in every component.

1. Automated Test Suite

All tests were run with pytest 9.2.0 on Python 3.12.10 (Windows). Result: 19 passed in 65.62 seconds.

Test file	Tests	Coverage
tests/test_api.py	4	System info, warm and cold-start recommendation endpoints, A/B metrics event
tests/test_cold_start.py	2	Popular diverse fallback, content-based onboarding fallback
tests/test_evaluation.py	5	NDCG (perfect and zero cases), Recall@K, MRR, A/B click handling
tests/test_ranking.py	2	DCN-v2 and Wide & Deep forward passes
tests/test_redis_cache.py	2	In-memory cache expiry, recommendation cache workflow
tests/test_retrieval.py	2	User and item tower output shapes, FAISS retriever
tests/test_spark_pipeline.py	2	Temporal split leakage check, feature integrity

2. Model Training

Stage 1: Two-Tower Retrieval

- Training data: 49,319 positive interactions (rating of 3.5 or higher)
- Loss: in-batch sampled softmax contrastive loss, temperature 0.07
- Embedding dimension: 64, L2-normalized
- Item catalog indexed in FAISS: 9,742 movies
- Validation loss: 4.7354 at epoch 1, 4.7350 at epoch 2 (see open items)

Stage 2: Deep Ranking Models

Both models were trained on chronological interaction records with negative sampling.

Model	Epochs	Training BCE	Validation BCE	Best validation ROC-AUC	Validation accuracy
DCN-v2 (Deep & Cross)	6	0.3620	0.3397	0.9188	85.76%
Wide & Deep	6	0.3868	0.3791	0.9045	84.04%

DCN-v2 achieved 1.43 percentage points higher validation ROC-AUC and higher accuracy.

3. Offline Recommendation Benchmark

Evaluated on the held-out temporal test set, split chronologically so that training timestamps precede validation and validation precedes test.

Configuration	Recall@50	NDCG@10	Hit Rate@10	MRR	Median latency
Stage 1: retrieval only (two-tower + FAISS)	0.0423	0.0149	16.67%	0.0626	3.28 ms
Stage 2: two-tower + Wide & Deep	0.0379	0.0178	22.22%	0.0537	2.55 ms
Stage 2: two-tower + DCN-v2	0.0332	0.0086	5.56%	0.0302	2.95 ms

Wide & Deep re-ranking raised NDCG@10 from 0.0149 to 0.0178 and Hit Rate@10 from 16.67% to 22.22% compared with retrieval alone. DCN-v2 re-ranking performed worse than retrieval alone on these top-rank metrics.

FAISS search over 9,742 items returning the top 100 candidates ran in under 1.5 ms per query on average.

4. Serving Latency

P50 and P95 across 500 simulated production requests.

Component	P50	P95	Target SLA
Redis cache check (hit)	0.78 ms	1.21 ms	5 ms
User tower inference (ONNX)	2.14 ms	3.56 ms	10 ms
FAISS MIPS search (top 100)	1.05 ms	1.89 ms	5 ms
Stage 2 deep ranker (100 items)	3.88 ms	5.42 ms	20 ms
Total, cold cache	8.24 ms	12.45 ms	50 ms
Total, cached	0.82 ms	1.25 ms	5 ms

All components and totals meet their targets.

5. Open Items and Limitations

These items should be resolved or disclosed before the report is submitted.

1. **Model ranking disagreement.** DCN-v2 has the higher validation AUC, but Wide & Deep is the better re-ranker on the offline test benchmark. The report treats the test benchmark as the decision metric for re-ranking, since it reflects the top-of-list quality users see.
2. **Small test set.** Hit Rate values (16.67% and 22.22%) are consistent with very few evaluated users. Relative lifts computed from so few hits are unstable and should be reported with the number of evaluated users and a confidence interval.
3. **Hit Rate denominators.** 16.67% corresponds to 1 of 6 users and 22.22% to 2 of 9 users, so the two figures may use different evaluation sets. Confirm the denominators before publishing the lift.
4. **Rounding of the NDCG lift.** 0.0149 to 0.0178 is a relative gain of about 19.5%, not 19.4%. Correct the figure or quote the unrounded values.
5. **Checkpoint selection.** The log marks epoch 1 as the best checkpoint, but epoch 2 has the lower validation loss (4.7350 versus 4.7354). Either the checkpoint logic or the logged values need correction.
6. **Stage 1 convergence.** Validation loss changed by only 0.0004 between epochs 1 and 2. The model may be under-trained, and more epochs or a learning-rate check would show whether retrieval quality can improve.
7. **Test module count.** The suite is described as covering 8 modules but lists 7 test files.

8. **Latency measurement scope.** Stage 2 median latency is 2.55 to 2.95 ms in the benchmark and 3.88 ms P50 in the serving breakdown. These come from different measurements and should be labeled as such.
9. **Presentation consistency.** The presentation describes FAISS retrieval as sub-millisecond, while the measured P50 is 1.05 ms. Use the measured figure.

6. Conclusion

The engine is functionally tested and meets its serving latency targets with margin. The offline benchmark shows a measurable ranking gain from Wide & Deep re-ranking. The DCN-v2 result is the main open question: it wins on validation AUC but not on top-rank metrics, and the test set is small. Resolving items 2 to 5 would make the benchmark claims defensible.